

Anveet Pal

Computer Science Engineer · Python Developer

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SUMMARY

Python developer building automation scripts, computer-vision modules, and data pipelines, with hands-on experience in backend development, REST APIs, async/webhook integrations, and ML model development.

SKILLS

Languages: Python, JavaScript, SQL.

Backend & Frameworks: FastAPI, Flask, Django, .NET, REST APIs

Frontend: React.js, Streamlit.

AI & ML: LLM tool-calling & agents (Ollama, OpenAI SDK), scikit-learn, XGBoost, distilBERT

Data & Analytics: Pandas, Seaborn, Power BI, Matplotlib, ETL/ELT

Computer Vision: OpenCV, YOLO (Ultralytics), ByteTrack, GPU/CPU inference (FP16)

Databases & Tools: MySQL, MSSQL, Git & GitHub, Postman, VS Code, PyCharm, Visual Studio 2022

Cloud & Deployment: Amazon EC2, Render, Vercel.

Others: Automation (Selenium), Playwright, CI/CD basics, C#, TypeScript, Next.js, Shopify, Figma, Microsoft Power Platform

EXPERIENCE

Python Developer · Miraigate Technologies

Feb 2026 – Present

- Built real-time video streaming pipelines in Python using LiveKit, publishing low-latency live streams to web clients for interactive real-time experiences.
- Designed async Python backend flows and webhook integrations to handle real-time data processing across multiple services.
- Developed computer-vision modules (OpenCV, YOLO) to process live video feeds for a real-time interactive gaming feature.
- Built and maintained Python automation scripts for multiple projects, adapting quickly to shifting priorities in a fast-paced startup environment.

Junior Consultant Intern · Hosho Digital

Jul 2025 – Nov 2025

- Developed and modified .NET backend system, managing databases and ensuring smooth frontend integration.
- Automated software tasks with Microsoft Power Platform.
- Performed data engineering and analysis with Pandas and Seaborn to deliver custom client solutions efficiently.

Python-ML Intern · Alveofit (Roundwork Technologies)

Feb 2025 – Mar 2025

- Built and deployed a two-stage XGBoost ML pipeline for real-time asthma severity prediction; monitored model performance through EDA.
- Implemented a Flask REST API with robust input validation and error handling, integrating the model for production use.

PROJECTS

Offline AI Coding Agent · Python · FastAPI · Ollama · LLM Tool-Calling

- Built a Claude-Code-style local coding agent from scratch — a plan→act→observe loop that orchestrates tools (reading, writing, and editing files, running commands), running fully offline on 8GB VRAM via Ollama (qwen3:8b / qwen2.5-coder:7b).
- Designed a swappable LLM abstraction layer supporting hybrid online/offline modes (local Ollama or OpenAI SDK), with JSON tool-call validation that auto-retries on malformed responses.
- Shipped three frontends on a single agent core — a Rich CLI, a FastAPI + WebSocket web UI, and a pywebview desktop app — bundled into a single Windows executable with PyInstaller.

AI vs Human Essay Classifier · Python · PyTorch · HuggingFace Transformers · scikit-learn

- Built an end-to-end NLP system to classify essays as human-written vs AI-generated, combining a fine-tuned DistilBERT model with interpretable classical-ML baselines on a ~74K-essay corpus.
- Fine-tuned DistilBERT to 0.90 F1 on a held-out test set, outperforming Random Forest (0.77) and Logistic Regression (0.74) stylometric baselines.
- Engineered a leakage-safe data pipeline (multi-dataset merge, deduplication, stratified 70/15/15 split) and trained on a consumer 8GB GPU (RTX 4060) using mixed-precision (bf16), dynamic padding, and gradient accumulation.
- Documented model limitations and responsible-use guidelines, including false-positive risk and bias against non-native English writers.

CCTV Vehicle Counter · Computer Vision · FastAPI · Python

- Real-time computer-vision pipeline that detects, tracks and counts vehicles from live CCTV streams with low-latency, GPU-accelerated inference (FP16).
- Designed a line-crossing counter with occlusion handling, ghost-track suppression, and multi-class aggregation across concurrent video streams (ByteTrack).

Group Coding-Practice Tracker · Django · DRF · React · PostgreSQL

- Built and deployed a full-stack web app (Django 5 + DRF, React 19 + TypeScript) on a fully free-tier stack (Render, Vercel, Neon Postgres), with a GitHub Actions CI pipeline.
- Implemented secure JWT authentication with refresh-token rotation, email verification, and a hardened password-reset flow.
- Designed a difficulty-weighted group leaderboard computed in a single database query using Django ORM annotations, cached in Redis with automatic invalidation on new submissions.
- Worked around free-tier limits by re-architecting the account-sync flow to fit Render's 60-second timeout and replacing Celery with a GitHub Actions cron job.

Disease & Asthma Prediction API · Flask · XGBoost · ETL

- ETL-driven Flask REST API serving an XGBoost model for real-time disease severity prediction with feature encoding and robust error handling.
- Source: [Link](#)

Rewards Management API · .NET 8 · C# · MySQL

- RESTful API for member accounts, reward-points accrual and coupon redemption, following clean-architecture principles.

EDUCATION

B.Tech, Computer Science & Engineering · Mediacaps University

2021 – 2025

CERTIFICATIONS

- Microsoft Power BI — Advanced Analytics, Data Visualization, Business Intelligence
- Alteryx — ETL, Data Analytics, Analytical Problem Solving
- Figma — UI/UX, Web Design